

SKILL WEEK 6

Task 1:

Code:

```
#include <stdio.h>

int main() {
    char input[200];
    int i;

    printf("Enter a string with escape characters: ");
    fgets(input, sizeof(input), stdin);

    printf("\nProcessed string:\n");
    for (i = 0; input[i] != '\0'; i++)
    {
        if (input[i] == '\\' && input[i + 1] !=
            '\0') {
            i++;
            switch (input[i])
            {
                case
                'n':
                    printf("\n");
                    break;
                case 't':
                    printf("\t");
                    break;
                case ' ':
                    printf(" ");
                    break;
                case '\\':
                    printf("\\");
                    break;
                case '$':
                    printf("$");
                    break;
                case '!':
                    printf("!");
                    break;
                default:
                    printf("%c", input[i]);
            }
        }
    }
}
```

```

    } else {          printf("%c",
input[i]);
    }
}

return 0;
}

```

```

skjuneeth@LAPTOP-D8SJ2A02:~$ nano skill6.c
skjuneeth@LAPTOP-D8SJ2A02:~$ gcc skill6.c -o skill6
skjuneeth@LAPTOP-D8SJ2A02:~$ ./skill6
Enter a string with escape characters: Hello\nWorld

Processed string:
Hello
World

```

Task 2:

Code:

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {

    pid_t pid;

    pid = fork();
    if (pid < 0) {
        perror("fork failed");    return
1;
    }

    if (pid == 0) {

        // Child process    printf("Child
process created.\n");    printf("Child
PID: %d\n", getpid());

        char *args[] = {"ls", "-l", NULL};    execvp(args[0], args);

        // This runs only if execvp fails
        perror("execvp failed");    exit(1);
    }
}

```

```
    } else {  
  
        // Parent process    printf("Parent  
process.\n");    printf("Parent PID:  
%d\n", getpid());  
  
        wait(NULL);  
  
        printf("Child process finished.\n");  
    }  
  
    return 0;  
}
```

Output:

```
skjuneeth@LAPTOP-D8SJ2A02:~$ nano skill6.1.c
skjuneeth@LAPTOP-D8SJ2A02:~$ gcc skill6.1.c -o skill6.1
skjuneeth@LAPTOP-D8SJ2A02:~$ ./skill6.1
Parent process.
Parent PID: 857
Child process created.
Child PID: 858
total 1008
-rwxr-xr-x 1 skjuneeth skjuneeth 16256 Sep 21 04:06 Experiment7
-rwxr-xr-x 1 skjuneeth skjuneeth 16304 Sep 21 04:43 Experiment7.1
-rw-r--r-- 1 skjuneeth skjuneeth 573 Sep 21 04:43 Experiment7.1.c
-rw-r--r-- 1 skjuneeth skjuneeth 756 Sep 21 04:06 Experiment7.c
-rwxr-xr-x 1 skjuneeth skjuneeth 16176 Sep 21 04:27 Experiment8
-rwxr-xr-x 1 skjuneeth skjuneeth 16296 Sep 21 04:31 Experiment8.1
-rw-r--r-- 1 skjuneeth skjuneeth 1397 Sep 21 04:30 Experiment8.1.c
-rw-r--r-- 1 skjuneeth skjuneeth 1176 Sep 21 04:27 Experiment8.c
drwxr-xr-x 11 skjuneeth skjuneeth 4096 Aug 20 05:04 OSPP
drwxr-xr-x 5 skjuneeth skjuneeth 4096 Aug 20 05:33 Team18_OSSP
-rw-r--r-- 1 skjuneeth skjuneeth 76 Jul 28 04:03 allow.c
-rwxr-xr-x 1 skjuneeth skjuneeth 16056 Sep 21 03:19 argpass
```